UIT2521 – Information Theory and Applications

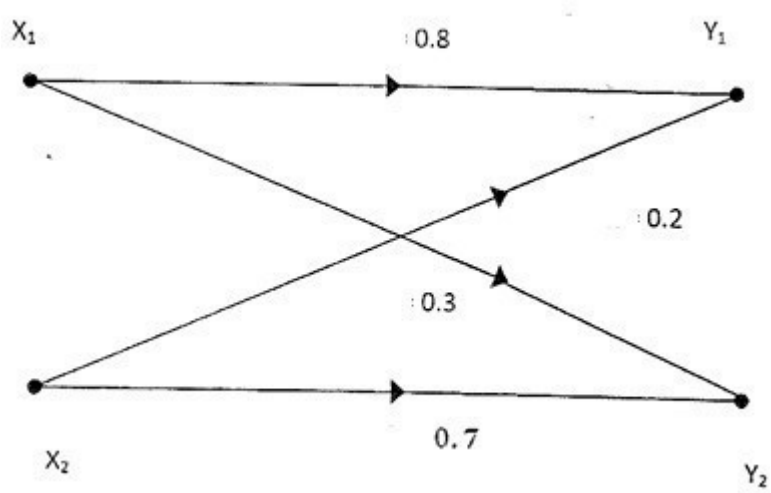
UNIT II INFORMATION THEORY FUNDAMENTALS

Tutorial – VI

Mutual Information

Date & Time: 04.08.2025 (Monday)

1. Find the mutual information for the discrete memoryless channel shown in figure. Given that $p(x_1) = 0.6$ and $p(x_2) = 0.4$.



Step 1. Source probabilities

$$P(X_1) = 0.6, \quad P(X_2) = 0.4$$

Step 2. Transition probabilities (from figure)

- From X_1:
  $P(Y_1|X_1) = 0.8, \; P(Y_2|X_1) = 0.2$
- From X_2:
  $P(Y_1|X_2) = 0.3, \; P(Y_2|X_2) = 0.7$

Step 3. Joint probabilities

$$P(X,Y) = P(X)P(Y|X)$$

- $P(X_1, Y_1) = 0.6 \times 0.8 = 0.48$
- $P(X_1, Y_2) = 0.6 \times 0.2 = 0.12$
- $P(X_2, Y_1) = 0.4 \times 0.3 = 0.12$
- $P(X_2, Y_2) = 0.4 \times 0.7 = 0.28$

✅ Check sum = 1.0

Step 4. Marginal probabilities of outputs

$$P(Y) = \sum_x P(X,Y)$$

- $P(Y_1) = 0.48 + 0.12 = 0.60$
- $P(Y_2) = 0.12 + 0.28 = 0.40$

Step 5. Entropies

(i) Entropy of output $H(Y)$

$$H(Y) = -\sum_y P(Y)\log_2 P(Y)$$

$$= -(0.6\log_2 0.6 + 0.4\log_2 0.4)$$

$$= -(0.6 \times -0.737 + 0.4 \times -1.322) = 0.971 \text{ bits}$$

(ii) Conditional entropy $H(Y|X)$

$$H(Y|X) = \sum_x P(x)H(Y|X=x)$$

- For X_1: $H(Y|X_1) = -(0.8\log_2 0.8 + 0.2\log_2 0.2)$
  $= -(0.8 \times -0.322 + 0.2 \times -2.322) = 0.722$
- For X_2: $H(Y|X_2) = -(0.3\log_2 0.3 + 0.7\log_2 0.7)$
  $= -(0.3 \times -1.737 + 0.7 \times -0.515) = 0.881$

Now weight:

$$H(Y|X) = 0.6(0.722) + 0.4(0.881) = 0.785$$

Step 6. Mutual Information

$$I(X;Y) = H(Y) - H(Y|X)$$

$$I(X;Y) = 0.971 - 0.785 = 0.186 \text{ bits}$$

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✅ **Interpretation:**

The channel transmits very little information about the input (0.19 bits). This means the output still has a lot of **uncertainty/noise** relative to the input.